FULL BASIS CHANGE

Same map, diagonal matrix

The transformation does not change, only its numerical description does.

BEFORE

Standard basis

$$A = \begin{pmatrix} 4 & 1 \\ 2 & 3 \end{pmatrix}$$

The map in standard coordinates

CHANGE TO THE EIGENVECTOR BASIS

put the eigenvectors in P, the eigenvalues in D

▼ P ON THE LEFT

AFTER

Eigenvector basis

$$D = \begin{pmatrix} 2 & 0 \\ 0 & 5 \end{pmatrix}$$

$A = PDP^{-1}$, with the eigenvectors in P

WHAT CHANGED The basis: in eigenvector coordinates the map only stretches, by 2 along (1, −2) and by 5

WHAT STAYED The transformation itself. To get A^{10}, raise only the diagonal entries: 2^{10} and 5^{10}.

along (1, 1).